

CREW MEMBERS / NAMES:

DATE / CLASS PERIOD:

CREW CALLSIGN / TEAM ID:

TESTED INSULATION CONFIGURATION:
[] A: Bare [] B: Cotton [] C: Mylar [] D: MLI

PART A. LIVE TELEMETRY DATA LOGGING TABLE (0 TO 15 MIN)

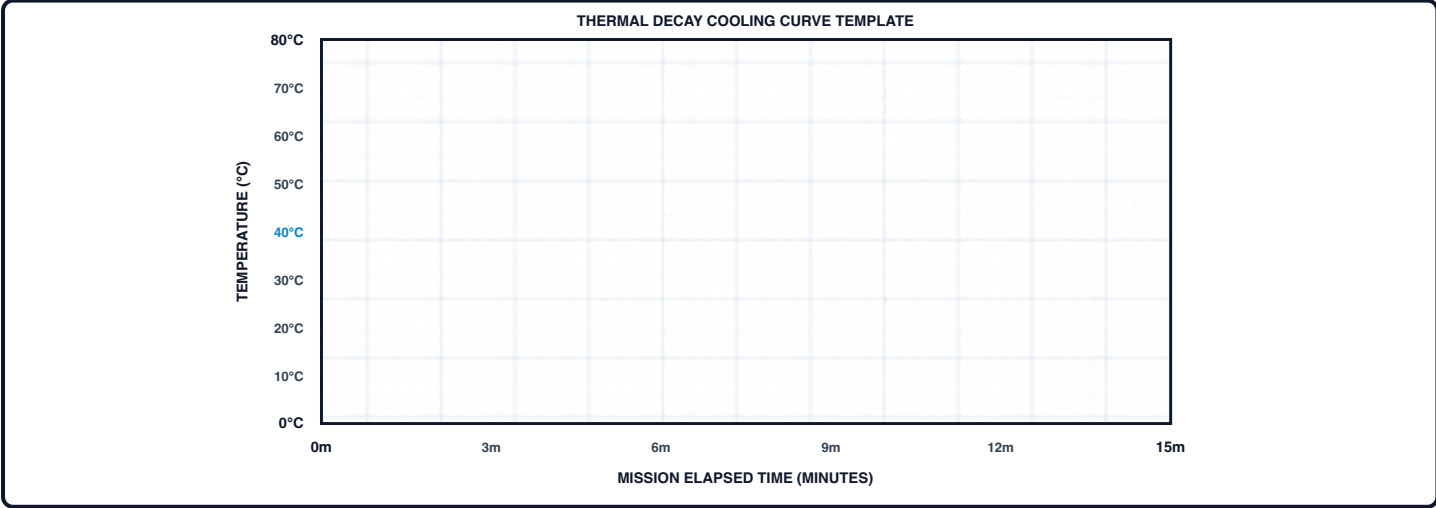
60s INTERVALS

TIME	PHYSICAL TEMP (°C)	PREDICTED TEMP (°C)	ABS DIFF ΔT (°C)	OBSERVER FIELD NOTES
0 min	80.0°C	80.0°C	0.0°C	Capsule submerged in 0°C ice bath
1 min				
2 min				
3 min				
4 min				
5 min				Check thermal equilibrium status
6 min				
7 min				
8 min				Mid-point verification mark
9 min				
10 min				
11 min				
12 min				
13 min				
14 min				
15 min				Final measurement checkpoint

PART B. THERMAL COOLING CURVE GRAPH GRID

PHYSICAL (•) VS PREDICTED (---)

Plot physical temperature readings as solid points (•) and join them. Plot theoretical model predictions as a dashed curve (---).



PART C. MANUAL THERMAL CORRELATION SCORE CALCULATION

PERFORMANCE EVALUATION

1. Sum of Absolute Differences:

Sum $|\Delta T| = \sum |T_{\text{phys}} - T_{\text{pred}}|$

2. Average Experimental Error:

Avg Error = Sum $|\Delta T|$ / 15

3. Digital Twin Correlation Score:

Score = 100 - (Avg Error × 5)

Sum $|\Delta T|$ = _____ °C

Avg Error = _____ °C

Score = _____ %

FINAL CREW RATING:

[] A+ (≥95%)

[] A (90-94%)

[] B (80-89%)

[] C (70-79%)

[] RE-CALIBRATE (<70%)

PART D. POST-FLIGHT ANALYSIS & REFLECTION QUESTIONS

6 KEY CONCEPT QUESTIONS

Q1: Thermal Radiation vs Conduction

Mylar reflects IR radiation in space. Why does Mylar alone perform poorly when submerged in an ice bath without cotton or bubble wrap?

Q2: Physical Meaning of Decay Rate Constant (k)

In Newton's Law $T(t) = T_{\text{env}} + (T_0 - T_{\text{env}}) \cdot e^{(-kt)}$, does a smaller k value indicate better or worse insulation? Explain using the formula.

Q3: Multi-Layer Insulation (MLI) Synergy

Explain how combining Cotton + Bubble Wrap + Mylar in Configuration D creates a synergistic defense against conduction, convection, and radiation.

Q4: Sources of Experimental Error

Identify two physical sources of error (e.g. sensor positioning, water leakage, cap sealing) that could cause readings to diverge from the model.

Q5: Environmental Boundary Shift

If your ice bath warms from 0°C to 4°C during the lab run, how should you adjust the simulation parameters in Mission Control to keep model validity?

Q6: Heat Capacity & Volume Scaling

If the core water volume is increased from 100ml to 200ml, how does total heat capacity ($Q = m \cdot c \cdot \Delta T$) affect the cooling rate k?

CREW LEAD SIGNATURE:

TEACHER VALIDATION:

FINAL GRADE: _____ / 100